

# CSE 3031 Computer Architecture

## Homework 1

Spring 2026

Deadline: Midnight of Thursday, Feb. 5

**Problem 1 (15 pts - 5/5/5):** Of the three factors in the processor performance equation ( $EX_{CPU} = IC \times CPI \times \text{cycle time}$ ), which is most influenced by:

- (a) The technology
- (b) The compiler
- (c) The computer architect

**Problem 2 (15 pts):** When running an integer benchmark on a RISC machine, the average instruction mix was as follows:

| Instructions | Average Frequency | Instructions   | Average Frequency |
|--------------|-------------------|----------------|-------------------|
| Load         | 26%               | Store          | 9%                |
| Arithmetic   | 14%               | Compare        | 13%               |
| Cond. branch | 16%               | Uncond. branch | 1%                |
| Call\return  | 2%                | Shift          | 4%                |
| Logical      | 9%                | Misc.          | 6%                |

The following measurements of average CPI for individual instruction categories were made:

| Instruction type     | Average CPI | Instruction type      | Average CPI |
|----------------------|-------------|-----------------------|-------------|
| All ALU instructions | 1           | Conditional branches: |             |
| Load-store           | 1.4         | — Taken               | 2.0         |
| Jumps                | 1.2         | — Not taken           | 1.5         |

Assume that 60% of the conditional branches are taken and that all instructions in the Misc. category are ALU instructions. What is the CPI of the benchmark on this RISC machine?

**Problem 3 (30 pts - 10/10/10):** Consider two implementations, M1 and M2, of the same ISA. We are interested in the performances of two programs, P1 and P2, which have the following instruction mixes:

| Operations | P1  | P2  |
|------------|-----|-----|
| Load/store | 40% | 50% |
| ALU        | 50% | 20% |
| Branches   | 10% | 30% |

The CPIs for each machine are:

| Operations | M1 | M2 |
|------------|----|----|
| Load/store | 2  | 2  |
| ALU        | 1  | 2  |
| Branches   | 3  | 2  |

- (a) Assume that the clock rate of M1 is 1 GHz. What should be the clock rate of M2 so that both machines have the same execution time for P1?
- (b) Assume now that both machines have the same clock rate and that P1 and P2 execute the same number of instructions. Which machine is faster for a workload consisting of equal runs of P1 and P2?
- (c) Find a workload (using only P1 and P2) that makes M1 and M2 have the same performance when they have the same clock rate.

**Problem 4 (20 pts - 10/5/5):** Assume the sequential execution occurs in 15% of the program, i.e. other 85% of the program can be parallelized with N processor with a speedup of N:

- (a) What is the maximum overall speedup with an infinite number of processors?
- (b) How many processors are required to be within 20% of the maximum speedup?
- (c) How many processors are required to be within 2% of the maximum speedup?

**Problem 5 (20 pts - 5/5/10):** Availability is the most important consideration for designing servers, followed closely by scalability and throughput.

- (a) We have a single processor with a failure in time (FIT) of 100. What is the mean time to failure (MTTF) for this system?
- (b) If it takes one day to get the system running again, what is the availability of the system?
- (c) Imagine that the government, to cut costs, is going to build a supercomputer out of inexpensive computers rather than expensive, reliable computers. What is the MTTF for a system with the 1000 processors mentioned in (a)? Assume the processor has an exponentially distributed lifetime and if one processor fails, the system fails.

**How to submit:** Write down your answer in a docx or pdf file. Upload the file through the submission link on Canvas.